



Project Case Study

Estimated time: 6 minutes

Capstone project case study: Merging data architectures of FashionMart and TrendyThreads

In this capstone project, you will act as an enterprise data architect at **TrendyThreads**, a rapidly growing digital apparel e-tailer. FashionMart, a traditional brick-and-mortar apparel retailer, has recently acquired your company. The acquisition creates a new combined entity, **FutureMart**, integrating FashionMart's streamlined data architecture with TrendyThreads' dynamic online infrastructure.

As the enterprise data architect, you are entrusted with ensuring the success of this integration. Your mission is to:

- Evaluate and analyze the existing data architectures of both companies.
- Design a cohesive and efficient data architecture for FutureMart that meets the needs of both the online and physical retail operations.
- Oversee data migration to the new unified architecture, ensuring minimal disruption.
- Integrate the data systems, facilitating seamless data flow across the newly combined entity.
- Draft a comprehensive data governance plan ensuring consistency, accuracy, and security in data management.
- Leverage the acquisition to improve other aspects of the enterprise architecture, enhancing overall operational efficiency.

Background: FashionMart & TrendyThreads

- **FashionMart:** A well-established brick-and-mortar apparel retailer with a strong presence in multiple cities. While it has decades of retail experience, its digital operations are underdeveloped, and it struggles to integrate physical and online sales effectively.
- **TrendyThreads:** A fast-growing online apparel e-tailer with a strong digital presence. It excels in personalized shopping experiences, AI-driven recommendations, real-time inventory tracking, and seamless online operations. However, given its startup origins, its policies and governance are not as mature.
- **The acquisition:** FashionMart acquired TrendyThreads to modernize its operations and strengthen its digital presence. The goal is to merge TrendyThreads' advanced technology with FashionMart's retail expertise, creating a unified, state-of-the-art enterprise architecture for the merged entity called FutureMart.

Capstone project: Phases and tasks to be completed

This project aims to **modernize FashionMart's operations** by leveraging **TrendyThreads' digital infrastructure**, ensuring seamless integration with FashionMart's physical retail operations. This will be accomplished through a structured, **phased migration approach**, covering data architecture evaluation, system integration, and governance.

Phase 1: Evaluate the data architecture of FashionMart & TrendyThreads

Tasks to be completed

- **Assess the existing data architectures** of both companies by examining the following:
 - Data Storage & Management
 - Data Integration
 - Data Quality & Consistency
 - Security & Compliance
 - Data Governance & Metadata
- **Identify inefficiencies, bottlenecks, and data silos** in the current systems
- **Analyze governance, security, and compliance measures** for both organizations

Phase 2: Design the data architecture of FutureMart

Tasks to be completed

- Use insights from Phase 1 to design an optimized data architecture for FutureMart
- Develop architectural diagrams showing the structure of the combined data ecosystem
- Define the new entity-relationship diagrams (ERDs) and table structures for the integrated databases
- Select appropriate databases and create initial schemas
- Document how the new architecture addresses inefficiencies and bottlenecks

Phase 3: Develop and implement a data migration strategy

Tasks to be completed

- Set up a staging environment for data migration testing
- Migrate data from RDBMS to NoSQL for unstructured data
- Migrate data from NoSQL to RDBMS where structured storage is needed
- Ensure data consistency, integrity, and transformation compatibility

Phase 4: Integrate data systems for seamless data flow

Tasks to be completed

- Develop ETL/ELT scripts to automate data extraction and transformation
- Create data pipelines for real-time data movement
- Design and optimize a data warehouse schema for reporting
- Develop a data pipeline for fraud detection models

Phase 5: Draft the data governance plan for FutureMart

Tasks to be completed

- Define governance objectives and scope aligned with business goals
- Establish a governance framework (centralized, decentralized, or federated)

- Assign governance roles (Data Owners, Stewards, Custodians)
- Develop policies and standards for data privacy, security, and quality
- Create data classification, access control, and metadata management policies

Final outcome

By the end of this capstone project, you will have successfully designed, migrated, integrated, and governed FutureMart's new data architecture, transforming it into a modern, seamless, omnichannel, and future-ready enterprise built on strong data foundations.